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Synergistic role of *Trichormus variabilis* and zeolites in three-layer culturing system for modulating the wastewater effluent community

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Due to their nitrogen-fixing capabilities, cyanobacteria hold significant potential for wastewater bioremediation through nutrient removal and modulation of the microbial community. The current study explored these traits using the cyanobacterium *Trichormus variabilis* strain AICB 1382 in combination with natural zeolites to treat municipal wastewater effluent. A combination of colorimetric, gravimetric, and 16 S/18S rDNA amplicon sequencing analyses was used to evaluate nutrient removal rates, biomass yield, and microbial community structure. The zeolites-AICB 1382 pair features (i.e. gradual release of nutrients by zeolites and vertical distribution of the cyanobacterium) enabled the stratification of the culturing system into three layers with distinct morphology and microbial populations. Results showed efficient removal of nitrate (up to 91.8%), ammonium (up to 97%), and phosphate (up to 99.2%), with enhanced biomass yields in zeolite-enriched cultures. *T. variabilis* reduced the diversity of the prokaryotic and eukaryotic community, lowering the presence of multidrug-resistant bacteria, whereas zeolites promoted the development of AICB 1382 and increased microbial diversity. The three-layer culturing system offers a promising solution for nutrient reclamation, biomass production, and pathogen reduction, with potential for scale-up as a semi-continuous, self-sustaining method that facilitates biomass harvesting while ensuring environmental safety for agricultural reuse or discharge into urban rivers.

Keywords *Trichormus variabilis*, Zeolites, Nutrients, Biomass, Prokaryotic community, Eukaryotic community

Cyanobacteria can thrive in various habitats, from dry lands[1] to various water sources, where they produce oxygen and regulate the nitrate/ammonium (NO_3^-/NH_4^+) : phosphate (PO_4^{3-}) ratio by nitrogen (N_2) fixation. N_2-fixing cyanobacteria were considered as alternatives to synthetic nitrogen fertilizers[2] whose usage resulted in significant environmental pollution[3] by soil acidification, humus decrement[4], groundwater eutrophication[3], and GHG (greenhouse gas) emissions[5]. Their efficacy lies in the efficiency of N_2-fixation that may reach up to 60 kg/ha/season of N_2 using *Anabaena* species[6] and their ability to tolerate various, even extreme conditions[7]. Cyanobacteria have been widely used in bioremediation[8], to reduce the levels of NH_4^+, NO_3^- (to synthesize proteins e.g., phycocyanin), PO_4^{3-} and to reduce the level of pathogenic bacteria[9]. Although recent studies have demonstrated the benefits of *Trichormus* cyanobacterial extracts in promoting plant growth[2], the authors emphasized the need for a phosphorus-enriched growth medium for *T. variabilis*. To address the natural scarcity and rapid depletion of phosphorus, they proposed utilizing wastewater as a phosphorus-rich alternative source.

One way to alleviate the need to supply nutrients involves culturing microalgae (cyanobacteria included) in piggery wastewater (WW)[10], municipal WW[11], and aquaculture WW[12]. Beyond WW organic load, cyanobacterial species were selected for traits that enhance bioremediation efficiency and facilitate biomass harvesting. Many cyanobacterial species (e.g. *Nostoc muscorum*, *Anabaena subcylindrica*, *A. oryzae*, *Spirulina*

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platensis, and *Geitlerinema* sp.) have been investigated for their potential use for reclamation of WW[8]. Different methods were used to tackle the harvesting process, either by using cyanobacteria rich in biopolymers (i.e. slime, sheath, and capsule)[13] that ease the formation of aggregates or by building cyanobacteria - trophic-related bacteria consortia[14,15]. Self-sustained consortia efficiently remove the biological oxygen demand (BOD) from the WW treatment plants (WWTP), clean up pollutants[16] and recover nutrients coupled with mitigation of CO_2[17]. This lowers the cost for aeration, which accounts for at least 50% of the energy inputs and expenses in biological treatment plants (TP)[18].

This study advances nutrient reclamation and cyanobacterial biomass production by developing a three-layer system combining *T. variabilis* AICB 1382, zeolites as substrate and the effluent from a municipal WWTP. Given that this effluent is currently discharged into an urban river, the study focused on characterizing microbial community dynamics, including the presence and reduction of pathogenic bacteria. *T. variabilis* (formerly *Anabaena variabilis*) was selected as a suitable candidate for this study owing to its tolerance to temperature fluctuations, its biofertilizer potential[2], and its capacity to grow in municipal WW environments[19]. The three-layer system was created by including natural zeolites, which are crystalline-hydrated aluminosilicates of alkaline and earth-alkaline elements (particularly of sodium and calcium). Due to their high capacity to exchange cations[20], the zeolites have been used for culturing cyanobacteria like *Arthrospira*[21], but also for remediation purposes[22]. The zeolites can adsorb cells[23] and can be used for EPS-producing bacteria immobilization[24]. The study aimed to investigate (i) the effect of *T. variabilis* on nutrient reclamation and its biomass productivity; (ii) the potential of *T. variabilis* to inhibit the growth of harmful bacteria containing multidrug resistance genes, which occur in the effluent[25], with the future aim of scaling the system before discharging the effluent into the river and, ii) the self-sustaining capacity of the culturing system based on its three-layer disposal. The analyses included the nutrient (NO_3^-/NH_4^+ and PO_4^{3-}) removal rate, the biomass yield, and the prokaryotic and eukaryotic community based on 16 S rDNA/18S rDNA amplicon sequencing.

Results

Three-layer culturing system – nutrient recovery and biomass productivity

Strain AICB 1382 formed buoyant, filamentous clusters in BG11 medium. In EZC system, the strain formed two distinct biomass layers separated by a clear effluent phase. The upper layer resembled a thick biofilm, composed of long, overlapping cyanobacterial filaments interspersed with bacteria, as observed by light microscopy. The lower layer on the zeolite surface appeared thin, homogeneous, and displayed an intense blue colour.

Nutrient recovery analysis revealed a generally higher rate for ammonium (NH_4^+) compared to nitrate (NO_3^-), with overall removal rate reaching 97% and 91.8%, respectively (Fig. S1A,B; Table S1). In experiments E1 and E2, treatments containing AICB 1382 (EC and EZC) consistently outperformed the zeolite-only control (EZ) in terms of nutrient recovery (Fig. S1A–C). Specifically, nitrate removal rates ranged from 17.4 to 76.4% in EZ, 80.1–88% in EC, and 67.1–91.8% in EZC. Ammonium removal followed a similar trend, ranging from 65.1 to 76.4% in EZ, 80.2–96.8% in EC, and 67.1–97% in EZC. Phosphate (PO_4^{3-}) removal was lowest in the EZ treatment (38.4–62.3%), increased substantially in EC (89.8–99.2%), and remained high in EZC (90.3–93.87%) (Fig. S1A–C; Table S1). These results underscore the synergistic effect of combining the cyanobacterium with zeolite, enhancing nutrient uptake and suggesting improved wastewater remediation potential.

Biomass productivity also varied depending on the effluent and treatment applied (Fig. S1D; Table S2). The highest biomass yield was consistently recorded in the system cultured with E2E effluent, reflecting the influence of effluent composition on cyanobacterial growth. Across all experimental conditions, the lowest biomass accumulation was observed in EZ, followed by EC, with the EZC treatment producing the highest yields. Overall biomass production ranged from 47.1 mg L^{-1} day^{-1} in EC to 156.2 mg L^{-1} day^{-1} in EZC, highlighting the significant contribution of both the cyanobacterium and the zeolite substrate to enhanced growth and potential for downstream biomass utilization (Fig. S1D; Table S2).

Analysis of the prokaryotic community

Composite samples analysis

The analysis of the composite samples collected by mixing all the layers showed that the treatments applied were one of the factors that shaped the taxa and their abundance in the prokaryotic community. The beta diversity analysis by PCoA (Fig. S2) matched the clusters with the treatments: effluent (E), EZ, and EC/EZC. The last two were separated in the UPGMA clustering (Fig. 1) which emphasizes sample similarity without reducing dimensions like PCoA.

Alpha-diversity assay strengthened and deepened this result showing differences in taxa occurrence, abundance, diversity, and dominance (Fig. S3). The number of taxa (Chao-1 index) differed between the effluents and the treatments, but their abundance and diversity (Evenness and Shannon-H indices) showed a similar pattern for the effluent (E) and the EZ condition (Fig. S4). No dominant taxa were found in these tanks, contrary to EC/EZC conditions. Comparison between the treatments pointed out a significant difference (ANOVA test) between the effluent (E) and the EZC samples based on the Shannon-H index ($F(3, 10) = 17.911$, $p < .001$) (Table S3). Levene's test indicated that the variances were homogenous, $F(3, 10) = 3.043$, $p = .079$; thus, Tukey's HSD Test for multiple comparisons showed that the mean value of the Shannon-H index was significantly different ($p < .001$, 95% C.I. = [1.550, 4.326]).

Another factor that influenced the structure of the prokaryotic community was the effluent type. Within the same cluster (EZ/EZC), the samples cultured in the first two effluents clustered separately from the third (Fig. 1), suggesting a separation due to the effluent microbial load. This fact was confirmed by the OTUs analysis (Fig. 2) where E1E and E2E shared 619 OTUs from 1389 to 1555 OTUs, making them more similar than the third effluent (482 OTUs). Even though their Chao-1 index was slightly different (Fig. S3), the first two effluents had

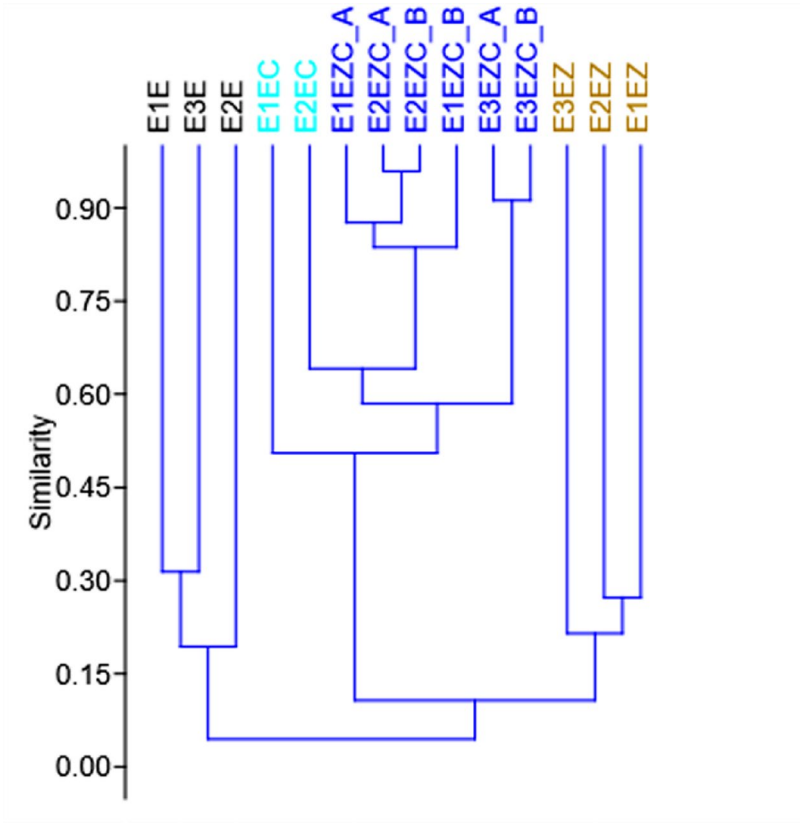

Fig. 1. UPGMA clustering of the biomass samples collected from the three experiments (E1, E2, E3) based on the OTUs abundance using the Bray-Curtis similarity matrix. Each experiment included the effluent (E), EZ, and 2 containers (A and B) of EZC. The experiments E1 and E2 also tested the EC condition.

equal OTU abundance (Evenness index). The Shannon-H index indicated a greater variety of species and a fairer distribution of individuals among species without dominant taxa in E1E and E2E than in E3E.

During culturing, the OTUs lowered in all tanks (Fig. 3A) retaining a common core for all samples (three-point and composite) (Fig. S5A-C) as follows: 16 OTUs in E1, 32 in E2, and 26 in E3 at the end of the experiments (Fig. S5A–C), regardless of culturing conditions. Thus, these OTUs were unresponsive to the conditions tested. Except for the common core or overlapping between two or more samples, each sample had a specific number of OTUs present (Fig. S5D–F).

Three-point samples analysis

Beyond the effects of culture conditions created by zeolites, strain AICB 1382, and the effluent composition, the microbial community exhibited variation among the three layers. When AICB 1382 was present (EC/EZC) the culturing system exhibited three distinctive layers. Layers 1 and 3 were significantly different from layer 2 according to the alpha diversity indices. The evenness (Fig. 3B), and the total diversity (Fig. 3D) indices were significantly larger in layer 2 relative to the evenness (ANOVA test $F(9, 26) = 13.670$, $p < .001$) and the diversity index (ANOVA test, $F(9, 26) = 15.694$, $p < .001$) of the samples collected from layers 1 and 3 (Tables S4, S5). Levene's test showed homogenous variances in both cases, $F(9, 26) = 2.498/1.070$, $p = .033/0.416$; thus, the Post-Hoc analyses using Tukey's HSD test showed a significant difference ($p < .001$) between ECZ1/ECZ3 and the rest of the samples.

The Dominance_D index (Fig. 3C) revealed that the microbial communities from layers 1 and 3 were dominated by a few taxa with high relative abundance. The SIMPER (Similarity Percentage) analysis outlined *T. variabilis* AICB 1382 among the top taxa that accounted for the differences among the samples (Table S6) and most probably was responsible with the large dominance index registered for layers 1 and 3. This taxon contributed the most to the overall dissimilarity among growth conditions (35.74%) from the top ten OTUs shown (49.27%). The variations in the relative abundance across layers supported the clustering patterns observed in the PCoA analysis (Fig. S6) and the UPGMA dendrogram (Fig. 4) which split the top and bottom layers from the middle layer for the EC/EZC conditions. For these treatments, middle-layer samples were partitioned by effluent type, with E1/E3 distinguished from E2.

Phylum-level analysis

Phylum-level analysis of composite samples revealed distinct variation in microbial composition (Figs. 5, S7). In the AICB 1382-systems (EC/EZC), several phyla - NB1-j, Cyanobacteria, Gemmatimonadota, Acidobacteriota,

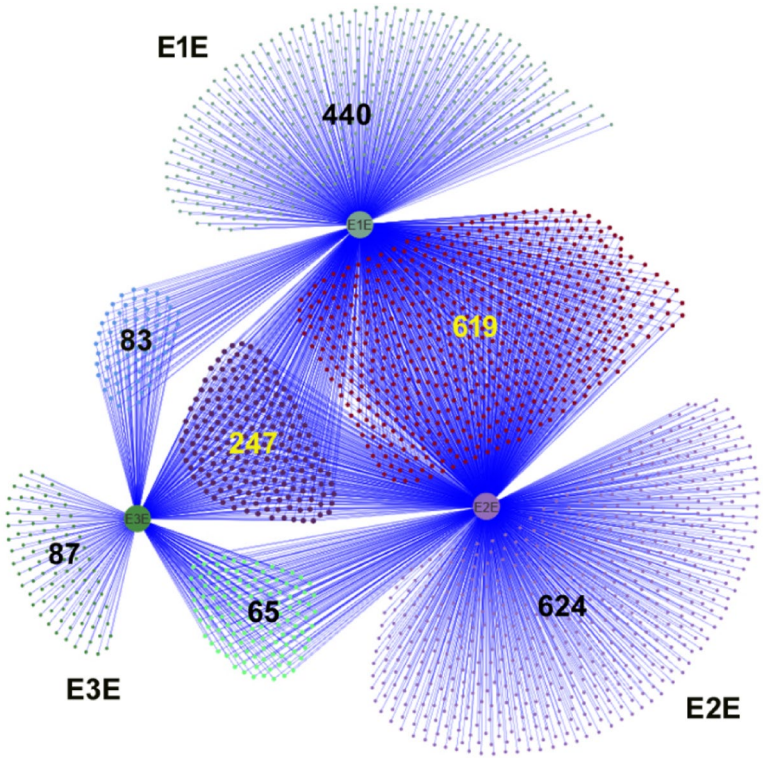

Fig. 2. The number of microbial OTUs and their common cores in the E1E, E2E, and E3E effluents. Each OTU was represented by a bullet. The numbers at the periphery indicate the unique OTUs specific to each effluent, while the overlapping areas represent the OTUs shared among the effluents.

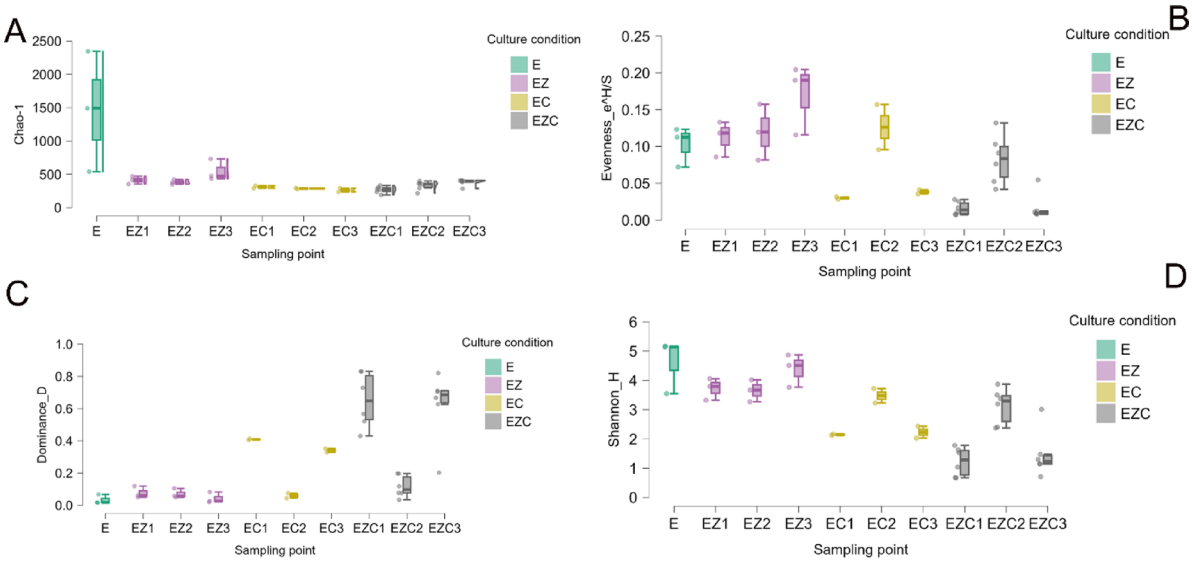

Fig. 3. The species richness (**A**), evenness (**B**), dominance (**C**), and diversity (**D**) indices based on the OTU abundances from different culturing conditions: EZ, EC, and EZC. The three-point samples were collected from the top (1), middle (2), and bottom (3) layers of the tanks. E = effluent.

Verrucomicrobiota, and Planctomycetota - were primarily observed. These were either underrepresented or absent in the effluent samples. Additional heterotrophic phyla such as Summerlaeota and WPS-2 (Eremiobacterota) were commonly associated. Deinococcota which appeared sporadically in the effluent, Dependentiae phylum, known for its intracellular lifestyle[26] and Patescibacteria characterized by minimal genomes[27] and epibiotic growth were better represented in AICB 1382 trials.

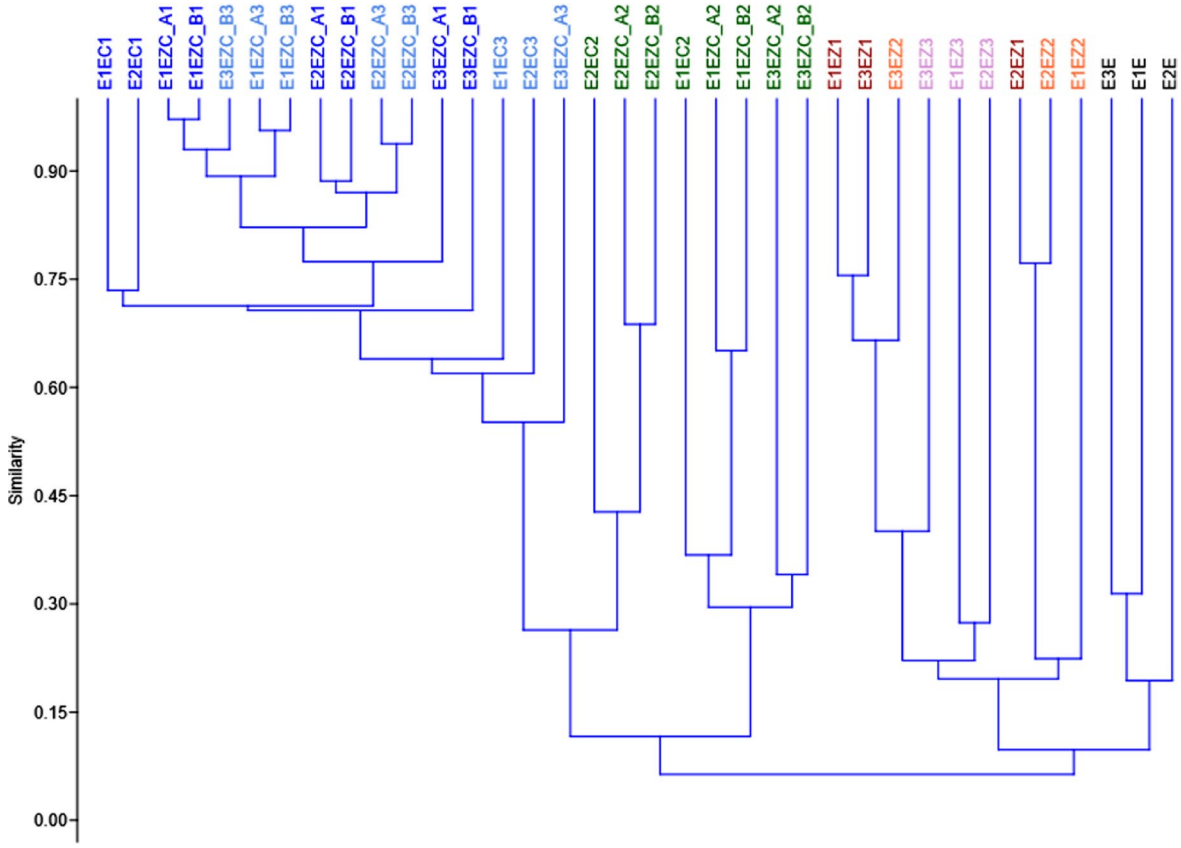

Fig. 4. UPGMA hierarchical clustering of the three-point samples (in different colors) based on the OTUs abundance using Bray-Curtis similarity matrix. Each experiment included the effluent (E), EZ, and 2 containers (A and B) of EZC. The experiments E1 and E2 included the EC condition.

Conversely, Proteobacteria (now *Pseudomonadota*), Bacteroidota, and Actinobacteriota were dominant across all treatments but were most abundant in effluent and EZ (zeolite only) conditions.

Approximately 50% of the phyla present in effluents were not detected after culturing. These lost taxa included several anaerobic and extremophilic groups such as Crenarchaeota, Euryarchaeota, Nanoarchaeota, Halobacteriota, Margulisbacteria, Elusimicrobiota, Desulfobacterota, and Fibrobacterota. Additionally, human-associated or potentially pathogenic families belonging to Proteobacteria, Actinobacteria, Campylobacterota, Fusobacteriota, Synergistota, Bacillota (formerly *Firmicutes*), and Spirochaetota were observed primarily in the effluents, but they were almost absent in the layers dominated by AICB 1382 (Fig. S8). Most genera detected showed abundances below 1% (e.g., *Enterobacter* spp., *Escherichia–Shigella* spp., *Rickettsia* spp., *Corynebacterium* spp., *Mycoplasma* spp., and *Lachnoclosterium* spp.). However, some genera such as *Closterium* spp., *Acinetobacter* spp., *Pseudomonas* spp., *Legionella* spp., *Aeromonas* spp., and *Mycobacterium* spp. accounted for at least 1% of the microbial community when present. The largest values were encountered for *Legionella* spp. and *Clostridium* spp. (cca. 4%), and *Pseudomonas* spp. (cca. 18%). These genera were also identified in the same effluent in a previous study, where they were associated with a high prevalence of antibiotic resistance genes[25]. The presence of *Legionella* (7 OTUs), *Clostridium* (8 OTUs), *Pseudomonas* (10 OTUs), *Acinetobacter* (10 OTUs) and *Mycobacterium* (8 OTUs) genera in the WWs could represent a potential health risk once they enter the receiving rivers, as they are considered important waterborne pathogens[25]. The presence of the oceanic and hydrothermal vent-associated phylum SAR324[28] occurred in the effluent samples.

Analysis of the eukaryotic community

Composite samples analysis

The eukaryotic community composition was driven by the same factors, i.e. culture conditions and the effluent microbial load. Unlike the prokaryotic community, the structure of the eukaryotic community was affected more by the effluent than by the treatment applied. PCoA of the composite samples revealed only partial separation according to the applied treatment (Fig. S9). However, clustering analysis (Fig. 6) provided greater resolution, distinguishing the EC/EZC samples from the first and third experiments from those of the second experiment, suggesting a potential effluent-driven grouping. OTU composition showed differences between effluents, with E2E showing the highest diversity (200 OTUs), compared to E1E (102 OTUs) and E3E (25 OTUs) (Fig. S10). At the end of the experiments, the EZC trials retained from the effluent a greater number of taxa than EZ/EC

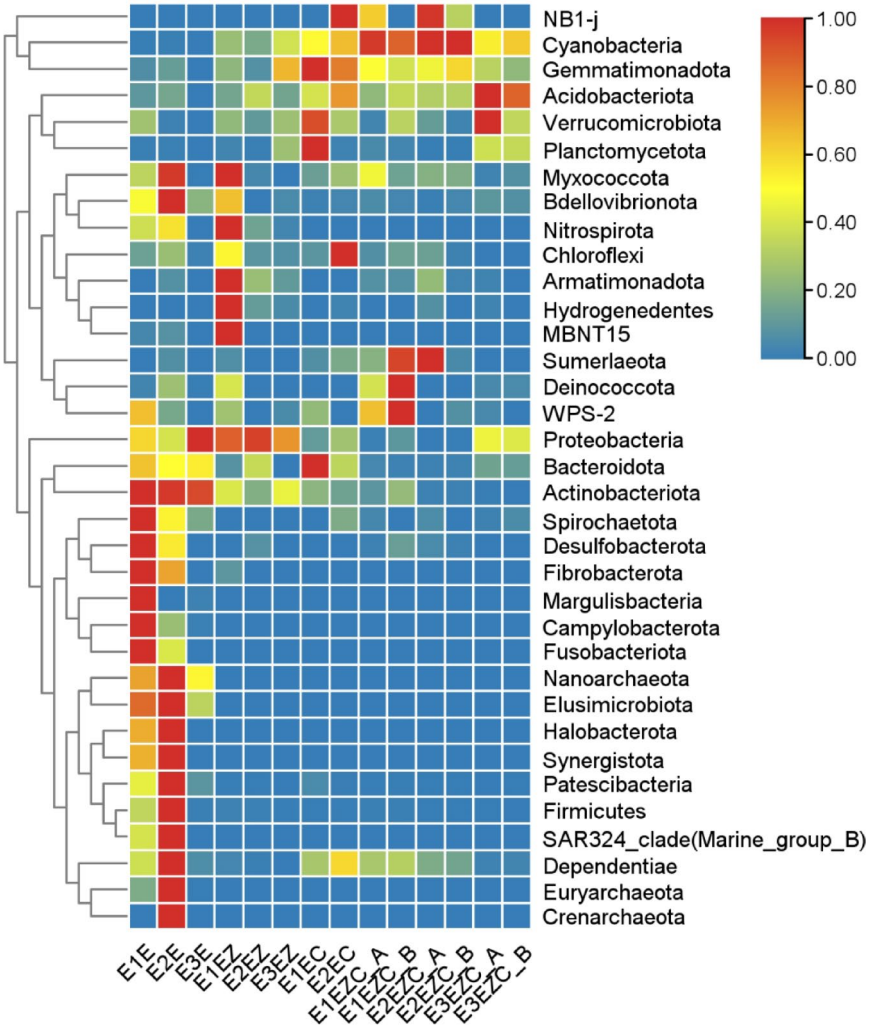

Fig. 5. Heatmap (scaled by row) of the first 35 phyla relative abundance in the total biomass collected from the effluent (E), EZ, EC, and EZC (tanks A and B) from E1, E2, and E3 experiments.

combinations (Fig. S11A–C). Similar to the prokaryotes, each sample harbored a set of distinct eukaryotic taxa (Fig. S11D–F).

Overall, alpha diversity indices showed a decline in the species richness during culturing and a similar pattern of the species diversity throughout all samples, except for E3EZC, where it slightly increased, probably due to the larger abundance (Fig. S12A–C). This condition exhibited the highest diversity and abundance indices, along with the lowest Dominance_D, indicating that effluent type played a key role in shaping the community.

Three-point samples analysis

The ANOVA assay of the samples collected from the three layers revealed no significant differences between groups ($p \geq .001$), regardless of experiment, culture condition, or sampling point (Fig. 7A–D). Beta-diversity assay by PCoA and UPGMA (Figs. 8, S13,) further demonstrated that clustering was influenced by the effluent composition and condition tested. In particular, all samples from the EZ trial clustered together; however, the AICB 1382–containing groups separated by effluent type, with E1 and E3 samples distinct from those of the E2 experiment (Fig. 8). The sampling point did not influence the grouping.

SIMPER analysis showed 93.94% average dissimilarity between the four main clades (Table S7). The top 10 OTUs accounted for 49.44% of overall dissimilarity, primarily including Bacillariophyceae, Xanthophyceae, Cryptomycota, and other heterotrophic taxa.

Phylum-level analysis

The analysis of the top 35 most abundant eukaryotic phyla in composite samples and across the three sampling points (Figs. 9, S13) revealed differences among the three effluents (E1E, E2E, and E3E) and culturing conditions (EZ, EC, and EZC). The E1E effluent was dominated by taxa belonging to Stramenopiles - frequently found in urban wastewater[29] and Cryptophyceae, known for thriving in diverse environments[30] Most of these taxa disappeared during culturing as well as the predators (by myzocystosis) (subphylum Protalveolata)[31] and

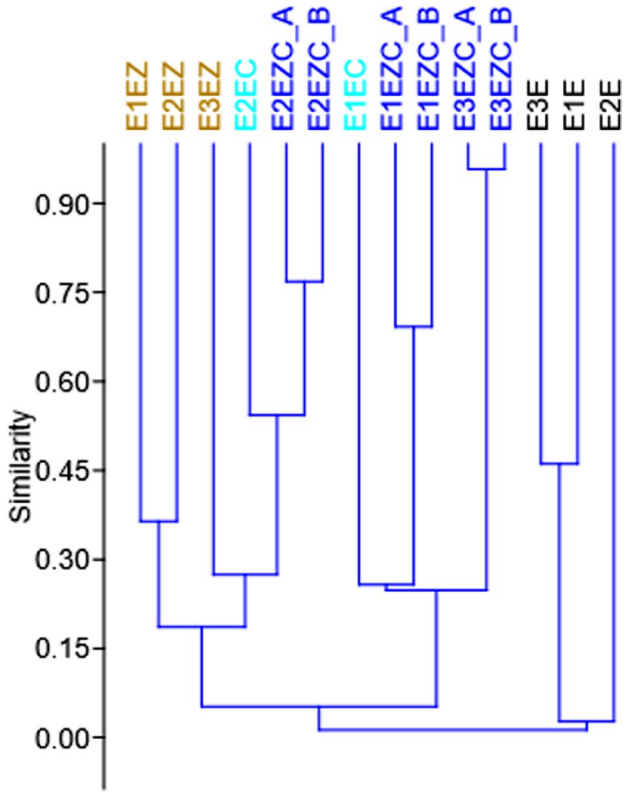

Fig. 6. UPGMA clustering of the biomass samples from the E1, E2, and E3 experiments based on the OTUs abundance, using Kulczynski distance. Each experiment included the effluent (E), EZ, and A and B duplicates of EZC. The E1 and E2 tested the EC condition.

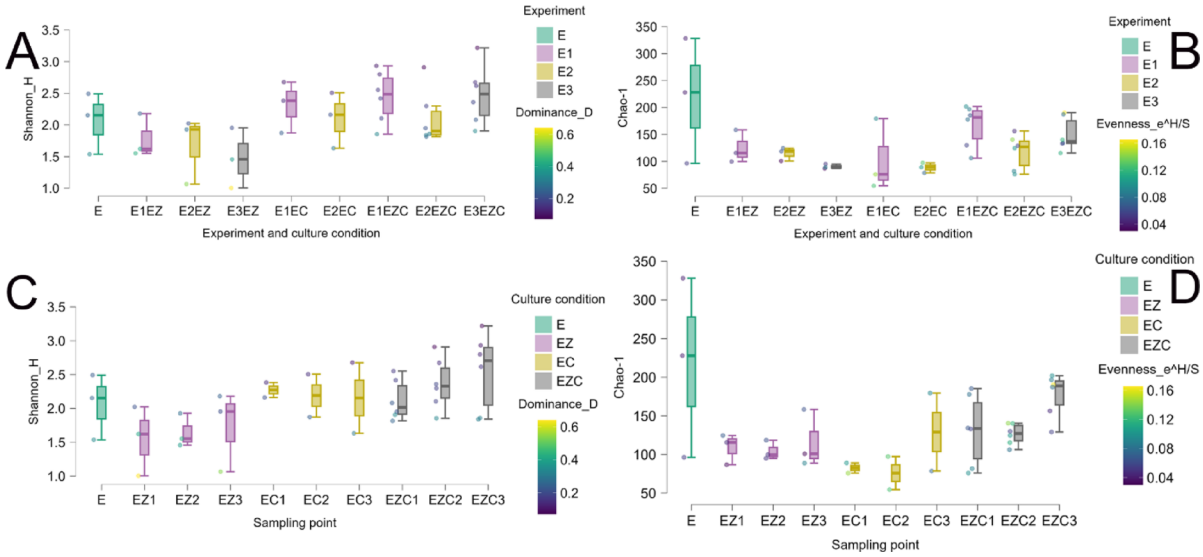

Fig. 7. The species richness, evenness, dominance and diversity indices based on the OTUs abundances from the eukaryotic samples. The samples were grouped by culture conditions EZ, EC, EZC and sampling points (top (1), middle (2) and the bottom layer (3)) (**C**, **D**), and by experiment (E1, E2, E3) (**A**, **B**). E = effluent samples. The colour of the bars represents group affiliation (shown in the upper right corner of each chart), while the gradient of the bullets indicates variation in the second variable analysed (shown in the lower right corner of each chart).

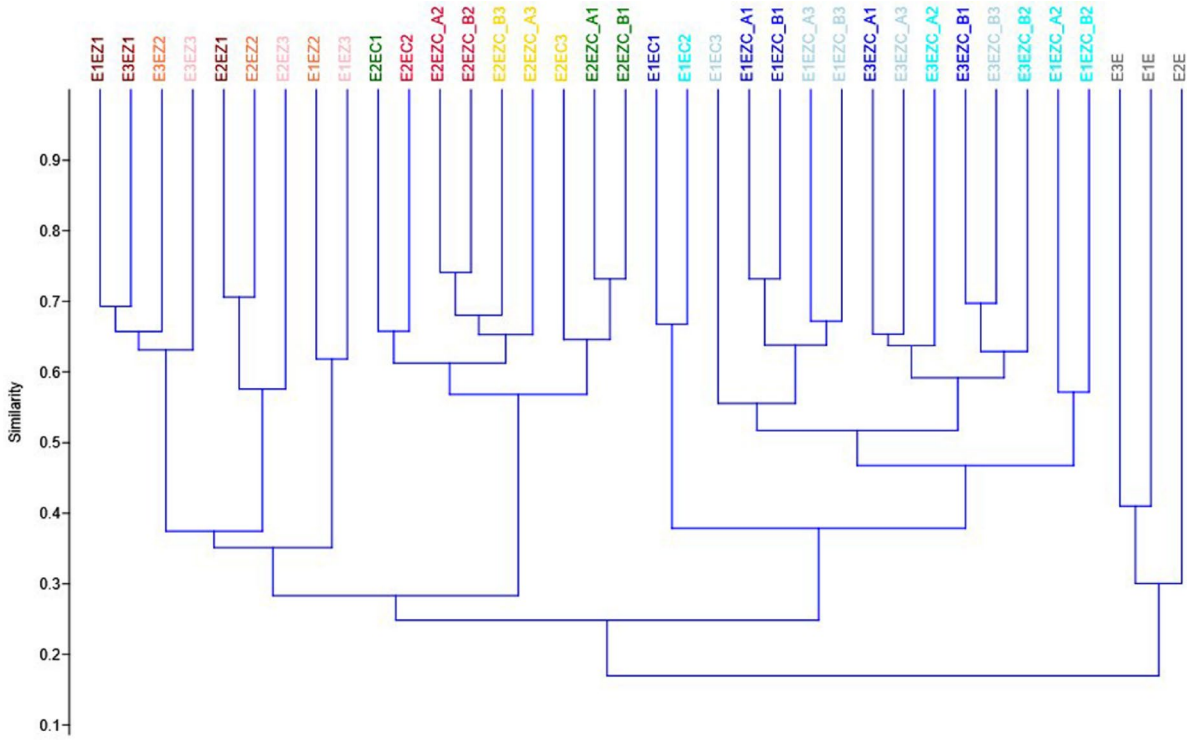

Fig. 8. UPGMA hierarchical clustering of the samples (colour-coded) collected from the three distinct layers: top (1), middle (2), and bottom (3), based on the OTUs abundance using Kulczynski distance. Each experiment included the effluent (E), EZ, and A and B duplicates of EZC. The E1 and E2 included the EC conditi.

anaerobic phagotrophs (MAST-12 group (Opalomonadea))[32] which did not persist post-cultivation. Similarly, in E2E, dominant metazoans like Annelida, Platyhelminthes, Mollusca, and Cnidaria, saprotrophs and parasites from Hyphochytridiomycota[33], soil fungi from Basidiomycota and LKM15[34], were lost during cultivation. None of these phyla, except for Protalveolata, were identified in the E3E eukaryotic community. This effluent stood out by its abundance of free-living protists from the Centrohelida phylum, found in most aquatic benthic environments where they feed on bacteria and other protists[35], and the species-rich Euglenozoa phylum, which contains free-living, parasitic, heterotrophic, and photosynthetic organisms[36].

An interesting observation was the presence of some taxa at the end of the experiments but their absence at the start in the corresponding effluents (i.e. E1E and E3E), such as Euglenozoa and Centrohelida phyla, in the E1EZ biomass, (Fig. 9), likely the result of sequencing limitations in detecting rare or low-abundance organisms[37]. Additionally, saprobic, chitin, and keratin-degrading chytrids from Chytridiomycota, which can occasionally act as parasites[38], were also identified. Although not present in the E3E community, taxa from the phyla Streptophyta (Viridiplantae), Phragmoplastophyta, and small meiobenthic worm- or cone-shaped animals from Gastrotricha that occur in high abundances in freshwater, marine, and brackish environments[39] were identified in high abundance in the EZ sample cultured in this effluent (Figs. 9, S13).

In the tested trials (EZ/EC/EZC), green algae (unassigned Chloroplastida) and diatoms (Bacillaryophyta) proliferated in all EZ samples regardless of the effluent type with co-occurring microbial phyla NB1-j, but they were sporadically present in the EC/EZC treatments. In EC/EZC cultures, the dominant phyla were primarily heterotrophs and decomposers. Dominant groups included Labyrinthulomycetes, typically marine saprotrophs or parasites[40], free-living heterotrophic protists from phylum Rigifilida[41], and fungi like Ascomycota and Cryptomycota[42].

E1EZC stood out due to its diverse and abundant eukaryotic community, including saprotrophs and parasites soil-fungi from Blastochlamidiomycota[43], plant-interacting fungi from Mucoromycota[44], molds that feed on bacteria from Fonticula[45], and amoeboid taxa like Heterolobosea, biflagellated protists from soil and aquatic habitats from Ancyromonadida[46], and free-living amoebae from soil and freshwater Nucleariidae[47]. These were joined by consumer phyla such as Cercozoa, Ciliophora, and Rotifera, commonly found in urban WW, particularly during warmer seasons[29].

The clustering analysis (Fig. S14) did not reveal a consistent grouping based on treatment or sampling point. The E2E effluent was more distantly placed compared to E1E and E3E, which intermixed with the cultured samples.

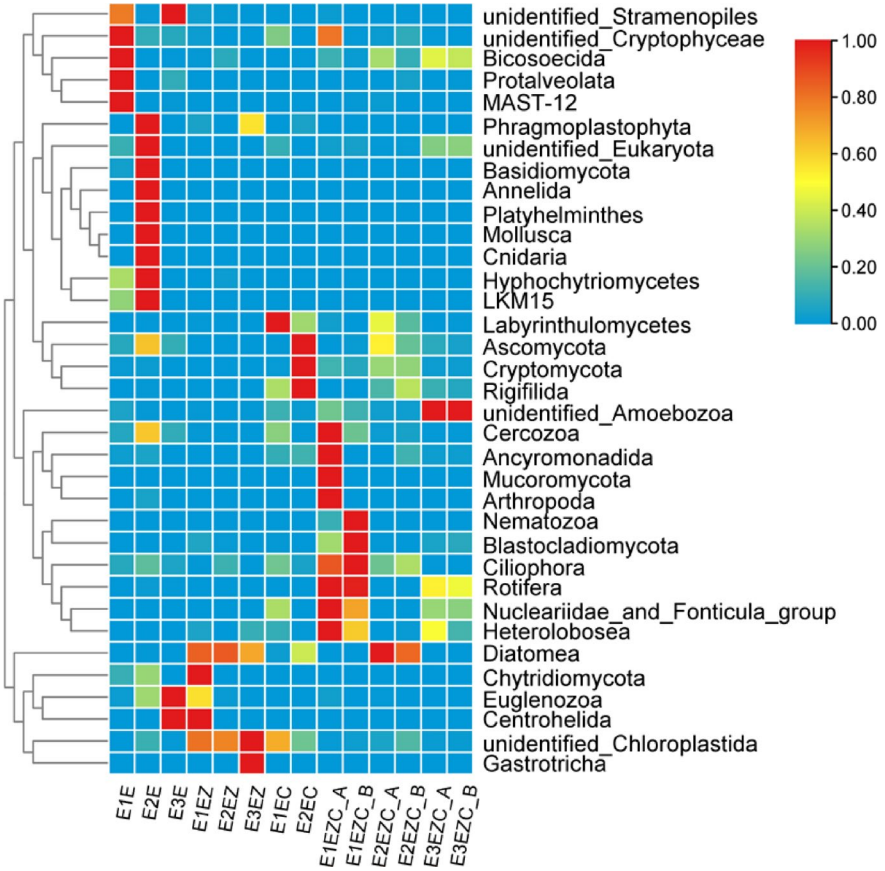

Fig. 9. Heatmap (scaled by row) of the first 35 eukaryotic phyla sorted according to their relative abundance in the biomass collected from the effluent (E), EZ, EC, and A and B duplicates of EZC. Data Availability. The 16 S rDNA and ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit (rbcL) gene sequences generated during the current study are available in GenBank database with the following IDs PV521982 (https://www.ncbi.nlm.nih.gov/nuccore/PV521982) and PV533918 (https://www.ncbi.nlm.nih.gov/nuccore/PV533918). The 16 S/18S rDNA amplicon datasets generated during the current study are available from the corresponding author on reasonable request. *Trichormus variabilis* AICB 1382 strain was deposited in the AICB Culture Collection and is available from the corresponding author on reasonable request.

Discussion

Community composition in engineered aquatic systems was shaped by interacting factors—zeolites, AICB 1382, and effluent type—causing spatial stratification. Due to its buoyant properties, *T. variabilis* AICB 1382 induced vertical stratification and formed two layers. The upper biofilm layer likely formed due to extracellular polymeric substances (EPS), which vary with conditions and microbial interactions[13]. Cyanobacterial EPS and oxygen promote bacterial growth, while bacterial metabolism supports algae via CO_2 release and organic matter breakdown[48]. The intense blue color in the EZC biomass suggests phycobiliprotein (phycocyanin, allophycocyanin) accumulation, sensitive to nitrogen[49]. Zeolites likely enhanced this by adsorbing and slowly releasing NH_4^+, reducing volatilization and maintaining nutrient supply[50]. This localized NH_4^+ may have supported the cyanobacterial layer at the zeolite surface, as NH_4^+ is energetically preferred over NO_3^-[51].

Effluent type and culturing conditions significantly impacted nutrient removal rate. EC/EZC systems showed the highest PO_4^{3-} and NO_3^-/NH_4^+ removal rate, while EZ performed best with E3E effluent, indicating an effluent-specific effect. Nitrogen recovery matched or exceeded reported rates for *Anabaena subcylindrica* (19.6–80%) and *Nostoc muscorum* (20.9–96%)[52–54]. Phosphate removal rate was also comparable or superior to values for *A. subcylindrica*, *N. muscorum* (50–81%), *Phormidium* sp. (62%)[55], and *Arthrospira* sp., which also reduced NH_4^+ from 100 mg L^{-1} to < 1 mg L^{-1}[56,57]. The enhanced performance in cyanobacteria-containing setups likely stemmed from consortia formation between AICB 1382 and native microbiota, known to boost nutrient recovery[58]. Although zeolite-enhanced systems yielded higher biomass, the observed productivity remains below potential levels reported in literature. For instance, *Anabaena* sp. reached 720 mg L^{-1} day^{-1} in synthetic medium and 400 mg L^{-1} day^{-1} in diluted pig slurry, with associated nitrogen removal rates of up to 2471 mg m^{-2} day^{-1}[59]. This suggests that nutrient limitation constrained the biomass accumulation. Future optimisation efforts should consider macronutrient supplementation or the use of nutrient-rich influents to achieve the full potential of the AICB 1382 strain.

Effluent composition had the strongest effect on eukaryotic communities, outweighing the influence of AICB 1382 or culturing design. Clustering analysis showed no consistent grouping by treatment or sampling stage, with E2E's distinct position highlighting effluent chemistry's role. The difference in the effluent's chemistry may be due to the seasonal and operational changes in WWTP which introduce variability in the microbial populations[60,61]. Nutrient stoichiometry, especially deviations from the Redfield N/P ratio of 16 (range: 8.2–45.0), strongly influences microbial diversity[62,63]. Nevertheless, core bacterial groups—Proteobacteria, Bacteroidota, and Actinobacteriota—were consistently present, reflecting their ecological importance and adaptability in freshwater systems[64]. Interestingly, SAR324—a bacterial phylum common in oceans, especially near hydrothermal vents—was detected in the effluent samples[28], likely originating from the activated sludge microbial community of the WWTP.

Zeolites acted as slow-release nutrient carriers and colonization surfaces, promoting both autotrophic and heterotrophic taxa. When combined with AICB 1382, they enhanced diversity, likely supporting rare or slow-growing species[65]. These effects varied with effluent and culture type, underscoring the need to align interventions with environmental conditions. Zeolites' aluminosilicate structure favored not only Bacillariophyta but other unassigned Chloroplastida growth in the EZ trial, though *T. variabilis* competition in EZC conditions likely reduced their abundance. The presence of NB1-j phyla[66] further supported diatom viability.

The spatial heterogeneity induced by AICB 1382 formed microzones favoring functionally distinct taxa[64], including bacteriochlorophyll-a and rhodopsin-bearing groups like Myxococcota, Chloroflexi, and Gemmatimonadota in cyanobacteria-rich layers[67,68]. Cyanobacteria-associated taxa like Summerlaeota and Eremiobacterota were enriched in cyanobacteria-containing systems[69,70], while the presence of resilient Deinococcota[71] underscored the selective pressures of engineered environments. N_2-fixing cyanobacteria also supported the growth of Verrucomicrobiota, Planctomycetota, and NB1-j, all key players in nutrient cycling and organic matter transformation[66,71,72].

EC/EZC trials enriched by *T. variabilis*' photosynthesis, supported diverse eukaryotic heterotrophs and decomposers commonly associated with primary producers. Elevated biodiversity in samples like E3EZC may reflect higher organic matter from intensified photosynthesis, fostering trophic complexity, or may result from native microbiota or effluent-specific nutrient profiles.

Several prokaryotic and eukaryotic phyla declined or disappeared, likely due to oxygenation, competition, or suppression by cyanobacterial metabolites[73], suggesting a possible sanitizing effect of the three-layer culturing systems. Lost taxa included extremophiles from salt lakes, intestines, anoxic sediments, and sludge digesters (e.g., Crenarchaeota, Euryarchaeota, Nanoarchaeota, Halobacteriota, Margulisbacteria, Elusimicrobiota, Desulfobacterota)[74–78], and gut-associated Fibrobacterota[79]. Most importantly, detected pathogens form Campylobacterota (formerly Epsilonproteobacteria)[80];, Fusobacteriota[81]; Synergistota, linked to human disease and found in WW, soil, and wells[82]; Bacillota (Firmicutes)[83]; and Spirochaetota, also[84] were not found at the end of the experiments. Genera like *Acinetobacter* (Proteobacteria, Moraxellaceae) and *Mycobacterium* (Actinobacteria) have been positively correlated with the prevalence of carbapenemase-encoding genes which are critical antibiotic resistance determinants[25]. The occurrence of potential pathogens in effluents underscores WW-related microbial risks, while their elimination in this three-layer system highlights its promise as a future biotechnological approach for WW sanitation and microbial risk mitigation.

Conclusion and perspectives

Most harmful bacterial phyla were reduced or eliminated after 14 days of culturing, likely due to oxygen exposure and the allelopathic effects of *T. variabilis* AICB 1382. This suggests the zeolite–AICB 1382 system poses minimal environmental risk for agricultural use or discharge into surface waters.

The AICB 1382 strain dominated EC/EZC treatments, reducing bacterial diversity, whereas zeolites helped maintain higher microbial diversity, particularly in the bottom layer. The three-point sampling revealed distinct microbial stratification, with culturing conditions having the strongest impact on community composition, followed by effluent properties. Zeolites facilitated spatial separation of AICB 1382, contributing to this stratification.

Overall, the culture system demonstrated that zeolites and AICB 1382 could modulate the eukaryotic community structure, but the effluent's chemical background largely dictated the trajectory and clustering of eukaryotic taxa.

In conclusion, microbial community dynamics in these systems emerge from the synergistic and antagonistic interplay between effluent characteristics, spatial configuration, and engineered interventions. Zeolites and *T. variabilis* served as both structural and biological agents capable of shaping ecological outcomes - from diversity reduction to functional specialization. This study demonstrates that combining stratification, nutrient modulation, and bioaugmentation can optimize microbial ecosystems for nutrient recovery, pathogen control, and ecological resilience. Harvested cyanobacterial biomass can be applied in agriculture or safely discharged, while the remaining biomass and zeolites can be reused to inoculate subsequent batches. Applications of the biomass on plants will be addressed in a forthcoming study."

Materials and methods

Strain selection and inoculum preparation

Thirty xenic cyanobacterial strains from the Algal and Cyanobacterial Culture Collection (AICB), Cluj-Napoca[85], were screened for N_2 fixation, growth in effluent, and cell aggregation. Axenic cultures were not used, as the non-sterile effluent and outdoor conditions make contamination unavoidable. Cultures were grown in nitrogen-free BG11 medium[86] under natural light (southern exposure) at 19 ± 2 °C. Biomass was sampled for DNA and microscopy, with two transfers into fresh medium to reach exponential growth before inoculation into

effluent. Strain AICB 1382 was selected based on its filament aggregation and dark blue color in effluent. Biomass was harvested (4000 rpm, 7 min), weighed, and used in experiments.

Light microscopy and taxonomic affiliation

Morphological analysis was done using light and fluorescence microscopy with a Nikon TE-2000 Eclipse microscope, and images were captured with a Nikon D90 camera. DNA was extracted using Quick-DNA™ Fecal/Soil Microbe Kits (Zymo Research, Irvine, CA, USA), following the manufacturer's protocol. PCR and sequencing targeted 16 S rDNA and rbcL genes using primers from Rudi et al.[87] and Frank et al.[88]. The PCR mix included 1.25 U DreamTaq DNA Polymerase (Fermentas, Canada), 1.5 mM $MgCl_2$, 0.2 mM dNTPs, and 0.4 µM primers in 50 µl total volume. Amplification was done with a Biometra TGradient cycler under standard conditions. Sequencing was performed by Macrogen Europe BV (The Netherlands), and sequences were deposited in GenBank[89] under IDs PV521982 and PV533918. Taxonomic identification as *Trichormus variabilis* was confirmed via BLAST search[89] against the GenBank Core Nucleotide database.

Experimental design and sampling assay

The 14-day experiment was performed using three separate effluent batches (E1E, E2E, E3E), resulting in three consecutive experimental runs (E1, E2, and E3). Three treatments were tested: effluent with zeolites (EZ), AICB 1382 cultured in effluent with zeolites (EZC), and AICB 1382 cultured in effluent without zeolites (EC). In the E3 run, the EC treatment was omitted due to logistical constraints. Each treatment was applied in a single container, except for EZC, which included two containers labelled A and B. Glass containers (30 × 19 × 20 cm) were kept at 23 ± 2 °C under a 16:8-h light/dark cycle with fluorescent light (25 µmol m^{-2} s^{-1}) and placed near a window (northern exposure) to enhance natural illumination. The final irradiance ranged from 40 to 50 µmol m^{-2} s^{-1}, varying with weather conditions (sunny versus cloudy days). No stirring was applied to prevent filament breakage[55]. Unfiltered effluent (1.780 L) originating from the activated sludge process of the municipal wastewater treatment plant in Cluj County, Romania, was collected in September (E1E), August (E2E), and July (E3E). Zeolites (271 g, 3–5 mm, Zeolites Production, Brașov, Romania) were added in EZ and EZC conditions, forming a 0.5 cm layer across 570 cm^2. AICB 1382 inoculum (500 mg wet biomass) was added to each condition; in EZC, it was mixed with zeolites and layered before pouring the effluent (4 cm liquid hight).

Sampling was performed in duplicate at both the beginning and the end of each experiment. At the start, 50 mL samples were collected from the three effluents (E1E, E2E, and E3E). At the end of each experiment, four samples were taken from each culture vessel; for this procedure, 50 mL were collected by pipetting separately from the upper and middle layers, without mixing them. For the bottom (zeolite) layer, an equivalent of 50 ml was estimated based on the total effluent volume (1.780 L) and zeolite weight (271 g), resulting in 7.5 g of zeolites. These were rinsed with 10 ml of a $MgSO_4 \cdot 7\ H_2O$ (10 mM) and Tween 80 (2000:1 v/v) solution to detach biofilm. After separate sampling, the contents were mixed thoroughly, and a final 50 ml composite sample was taken. All samples were filtered through sterile 0.22 µm cellulose nitrate membranes (Sartorius); filtrates were reserved for nutrient analysis. Filters were weighed before and after filtration to determine biomass, then stored at − 20 °C.

Biomass and nutrient analysis

Wet biomass yield was calculated by summing biomass from all sampling points and subtracting the 500 mg inoculum. Nutrient levels were measured using HANNA Instruments kits with a HAN I83399 Multiparameter Photometer (HANNA Instruments, Germany): PO_4^{3-} (kit HI 93713-01), and NO_3^-/NH_4^+ (kit HI 93767 A-50), reported in mg L^{-1}.

16 S/18S rRNA gene metagenomic sequencing

Nucleic acids were extracted from membranes using the kit described in Sect. 5.2, with duplicates pooled. PCR, quality control, amplicon library preparation, and sequencing were performed by Novogene CO using Illumina PE250 (30 K tags/sample). The bacterial 16 S rRNA V3–V4 region was amplified with primers 341 F/806R[90], and the eukaryotic 18 S rRNA V4 region with primers 528 F/706R[91]. Reads were demultiplexed, barcodes/primers removed, and merged with FLASH[92]. Quality filtering followed QIIME 2[93]; chimeras were removed using UCHIME. OTUs were clustered at 97% similarity via UPARSE[94] Taxonomic classification used Mothur (archaea/bacteria) and RDP (eukaryotes) with the SILVA database v138.1[95].

Diversity, statistical analysis, and visual representation of taxa

To reduce experimental error and ensure comparability, OTU abundances were normalized to the sample with the fewest sequences. Alpha diversity was assessed using Chao-1, Shannon_H, Evenness e^H/S, and Dominance_D indices. Beta diversity was analyzed via SIMPER, PCoA, and UPGMA clustering using Bray–Curtis and Kulczynski distances. Analyses were done in PAST 4.13[96], and one-way ANOVA was performed in JASP v.0.19.3 (2025). OTU visualization via Venn and flower plots used EVenn[97], and heatmaps were created in TBtools-II[98].

Data availability

The 16 S rDNA and ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit (rbcL) gene sequences generated during the current study are available in GenBank database with the following IDs PV521982 ([https://www.ncbi.nlm.nih.gov/nuccore/PV521982](https:/www.ncbi.nlm.nih.gov/nuccore/PV521982)) and PV533918 ([https://www.ncbi.nlm.nih.gov/nuccore/PV533918](https:/www.ncbi.nlm.nih.gov/nuccore/PV533918)).The 16 S/18S rDNA amplicon datasets generated during the current study are available from the corresponding author on reasonable request.*Trichormus variabilis* AICB 1382 strain was deposited in the AICB Culture Collection and is available from the corresponding author on reasonable request.

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Declarations

Competing interests
The authors declare no competing interests.

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